CMSRN performance under increasing routine-pattern deviation

H6 (30 min) · full test set · training-fitted scenario thresholds

(a) Chengdu · inflow

	Low	Moderate	High	Extreme
HA	+85.1	+62.8	+20.2	+12.0
SVR	+51.0	+31.2	+11.9	+8.0
VAR	+59.2	+39.8	+17.1	+8.1
DCRNN	+60.8	+17.5	+4.9	+5.0
STGCN	-1.1	+1.0	+1.2	+2.7
GWNet	-1.0	+2.4	+2.2	+3.7
STNorm	-4.5	-1.2	-1.2	+0.4
DSTAGNN	+81.9	+58.5	+20.5	+5.3
PDFormer	+78.5	+64.1	+52.1	+58.4
STWave	-4.9	-2.0	-1.5	+0.4
PatchTST	+70.0	+12.5	+7.5	+9.8
TimeMixer	+71.1	+17.3	+10.5	+9.9
STDN	+85.5	+70.4	+32.6	+5.6
MetaDG	+84.7	+66.5	+33.1	+12.4
HyperD	-0.7	+0.7	+0.2	+3.3
RAST	-0.7	+1.8	+2.5	+4.6
CMSRN MAE	**0.161**	**0.344**	**0.705**	**1.340**
Valid n	59,917	129,101	182,079	128,448

(b) Chengdu · outflow

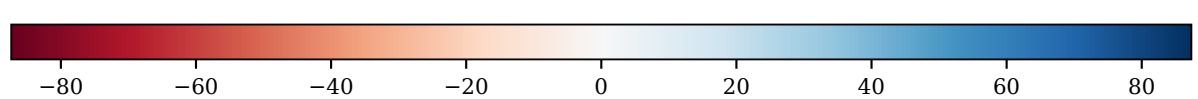
	Low	Moderate	High	Extreme
HA	+86.8	+58.5	+15.4	+9.0
SVR	+53.5	+25.7	+8.8	+5.4
VAR	+62.3	+33.2	+12.2	+6.0
DCRNN	+59.9	+12.1	+2.3	+2.1
STGCN	-1.2	-0.3	+0.2	+1.7
GWNet	-2.3	+0.4	+0.5	+1.9
STNorm	-2.8	-1.8	-1.5	-0.2
DSTAGNN	+84.8	+55.6	+18.5	+4.6
PDFormer	+80.3	+61.2	+50.5	+54.7
STWave	-3.1	-2.2	-1.2	-0.6
PatchTST	+72.3	+9.1	+5.1	+7.8
TimeMixer	+73.2	+14.0	+8.3	+8.4
STDN	+87.3	+66.3	+25.9	+4.3
MetaDG	+86.7	+63.2	+28.8	+10.2
HyperD	-1.1	-0.6	-0.3	+1.1
RAST	-0.9	+1.1	+1.4	+2.7
CMSRN MAE	**0.139**	**0.380**	**0.765**	**1.246**
Valid n	58,257	150,159	182,619	135,869

(c) Shanghai · inflow

	Low	Moderate	High	Extreme
HA	+82.7	+39.2	+12.3	+13.3
SVR	+50.1	+17.8	+8.4	+7.7
VAR	+60.5	+25.3	+10.8	+7.8
DCRNN	+10.1	+2.2	+1.9	+3.6
STGCN	+2.2	+1.2	+1.4	+4.0
GWNet	+1.2	+1.1	+1.6	+3.6
STNorm	-0.1	-1.1	-1.2	+0.5
DSTAGNN	+78.7	+33.0	+8.9	+5.1
PDFormer	+74.5	+52.5	+48.8	+57.7
STWave	-0.9	-1.0	-1.1	+0.2
PatchTST	+57.9	+6.4	+5.4	+10.2
TimeMixer	+58.9	+10.6	+8.2	+10.2
STDN	+84.6	+53.9	+25.6	+15.3
MetaDG	+83.1	+49.8	+24.8	+15.2
HyperD	+5.0	+0.8	+0.3	+2.8
RAST	+1.0	+0.9	+1.4	+3.1
CMSRN MAE	**0.177**	**0.536**	**0.851**	**1.296**
Valid n	92,199	223,697	222,486	115,210

(d) Shanghai · outflow

	Low	Moderate	High	Extreme
HA	+84.4	+37.9	+12.1	+12.1
SVR	+53.6	+16.5	+8.1	+7.2
VAR	+64.9	+23.6	+10.8	+7.4
DCRNN	+11.1	+2.1	+1.6	+3.4
STGCN	+2.1	+0.7	+1.1	+3.0
GWNet	+0.8	+0.6	+1.1	+3.6
STNorm	+0.1	-1.2	-1.2	+0.4
DSTAGNN	+81.2	+32.1	+9.3	+3.6
PDFormer	+76.7	+52.2	+49.9	+58.8
STWave	-0.3	-1.1	-1.0	+0.2
PatchTST	+61.5	+6.0	+5.1	+9.5
TimeMixer	+62.2	+9.8	+8.1	+9.7
STDN	+85.7	+43.0	+14.0	+6.1
MetaDG	+84.8	+49.6	+28.0	+16.6
HyperD	+1.9	+0.4	+0.0	+1.1
RAST	+2.3	+1.2	+1.5	+4.1
CMSRN MAE	**0.158**	**0.553**	**0.870**	**1.480**
Valid n	84,381	216,920	227,734	110,618



CMSRN MAE improvement over each baseline (%)

Improvement = 100 × (MAEbaseline − MAECMSRN) / MAEbaseline. Blue: CMSRN better; red: baseline better.

All finite values are shown without clipping; gray cells indicate an empty or undefined group.